DBMS Mini Project

Blockchain Transaction Explorer & Analytics Platform

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1. DML – Data Insertion

```
PES2UG24CS646_PES2UG24CS916>INSERT INTO users (name, email, role) VALUES
    -> ('Arjun Sharma', 'arjun.sharma@gmail.com', 'Admin'),
    -> ('Rahul Kumar', 'rahul.kumar@gmail.com', 'Analyst'),
    -> ('Sneha Rao', 'sneha.rao@gmail.com', 'User'),
    -> ('Aditya Singh', 'aditya.singh@gmail.com', 'User'),
    -> ('Priya Nair', 'priya.nair@gmail.com', 'Analyst'),
    -> ('Karan Patel', 'karan.patel@gmail.com', 'User'),
    -> ('Ananya Das', 'ananya.das@gmail.com', 'User'),
    -> ('Rohit Verma', 'rohit.verma@gmail.com', 'Analyst'),
    -> ('Meera Iyer', 'meera.iyer@gmail.com', 'User'),
    -> ('Vikram Joshi', 'vikram.joshi@gmail.com', 'Admin');
Query OK, 10 rows affected (0.01 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

```
PES2UG24CS646_PES2UG24CS916>INSERT INTO address
    -> (address, address_type, current_balance, last_seen)
    -> VALUES
    -> ('0x1111111111111111111111111111111111111111',
    ->  'EOA', 12.5000000000, '2026-09-01 10:00:00'),
    ->
    -> ('0x2222222222222222222222222222222222222222',
    ->  'EOA', 25.7500000000, '2026-09-02 11:00:00'),
    ->
    -> ('0x3333333333333333333333333333333333333333',
    ->  'Contract', 100.0000000000, '2026-09-03 12:00:00'),
    ->
    -> ('0x4444444444444444444444444444444444444444',
    ->  'EOA', 8.2500000000, '2026-09-04 13:00:00'),
    ->
    -> ('0x5555555555555555555555555555555555555555',
    ->  'EOA', 50.0000000000, '2026-09-05 14:00:00'),
    ->
    -> ('0x6666666666666666666666666666666666666666',
    ->  'Contract', 75.5000000000, '2026-09-06 15:00:00'),
    ->
    -> ('0x7777777777777777777777777777777777777777',
    ->  'EOA', 18.7500000000, '2026-09-07 16:00:00'),
    ->
    -> ('0x8888888888888888888888888888888888888888',
    ->  'EOA', 32.0000000000, '2026-09-08 17:00:00'),
    ->
    -> ('0x9999999999999999999999999999999999999999',
    ->  'Contract', 200.0000000000, '2026-09-09 18:00:00'),
    ->
    -> ('0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa',
    ->  'EOA', 5.5000000000, '2026-09-10 19:00:00'),
    ->
    -> ('0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb',
    ->  'EOA', 45.0000000000, '2026-09-11 10:00:00'),
    ->
    -> ('0xcccccccccccccccccccccccccccccccccccccccc',
    ->  'Contract', 150.0000000000, '2026-09-12 11:00:00');
Query OK, 12 rows affected (0.00 sec)
Records: 12  Duplicates: 0  Warnings: 0
```

```
PES2UG24CS646_PES2UG24CS916>INSERT INTO block
    -> (block_number, block_hash, timestamp, transaction_count, gas_used)
    -> VALUES
    -> (100001,
    ->  '0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa',
    ->  '2026-09-01 10:00:00', 2, 210000),
    ->
    -> (100002,
    ->  '0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb',
    ->  '2026-09-02 11:00:00', 2, 280000),
    ->
    -> (100003,
    ->  '0xccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccc',
    ->  '2026-09-03 12:00:00', 2, 150000),
    ->
    -> (100004,
    ->  '0xddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddd',
    ->  '2026-09-04 13:00:00', 2, 350000),
    ->
    -> (100005,
    ->  '0xeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeee',
    ->  '2026-09-05 14:00:00', 2, 420000),
    ->
    -> (100006,
    ->  '0xffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffff',
    ->  '2026-09-06 15:00:00', 2, 230000),
    ->
    -> (100007,
    ->  '0x1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef',
    ->  '2026-09-07 16:00:00', 2, 300000),
    ->
    -> (100008,
    ->  '0xabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcd',
    ->  '2026-09-08 17:00:00', 2, 170000),
    ->
    -> (100009,
    ->  '0x9876543210abcdef9876543210abcdef9876543210abcdef9876543210abcdef',
    ->  '2026-09-09 18:00:00', 2, 390000),
    ->
    -> (100010,
    ->  '0xfedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbabc',
    ->  '2026-09-10 19:00:00', 2, 250000);
Query OK, 10 rows affected (0.00 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

```
PES2UG24CS646_PES2UG24CS916>INSERT INTO watchlist
    -> (name, user_id)
    -> VALUES
    -> ('High Value Wallets', 1),
    -> ('DeFi Addresses', 2),
    -> ('Whale Tracking', 3),
    -> ('Exchange Wallets', 4),
    -> ('Smart Contracts', 5),
    -> ('Suspicious Addresses', 6),
    -> ('Token Monitoring', 7),
    -> ('Daily Transactions', 8),
    -> ('Large Transfers', 9),
    -> ('Research Wallets', 10);
Query OK, 10 rows affected (0.00 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

```
PES2UG24CS646_PES2UG24CS916>INSERT INTO watchlist_address
    -> (watchlist_id, address)
    -> VALUES
    -> (1, '0x1111111111111111111111111111111111111111'),
    -> (1, '0x5555555555555555555555555555555555555555'),
    ->
    -> (2, '0x3333333333333333333333333333333333333333'),
    -> (2, '0x6666666666666666666666666666666666666666'),
    ->
    -> (3, '0x9999999999999999999999999999999999999999'),
    -> (3, '0xccccccccccccccccccccccccccccccccccccccc'),
    ->
    -> (4, '0x2222222222222222222222222222222222222222'),
    -> (4, '0x8888888888888888888888888888888888888888'),
    ->
    -> (5, '0x3333333333333333333333333333333333333333'),
    -> (5, '0x9999999999999999999999999999999999999999'),
    ->
    -> (6, '0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa'),
    ->
    -> (7, '0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb'),
    ->
    -> (8, '0x7777777777777777777777777777777777777777'),
    ->
    -> (9, '0x8888888888888888888888888888888888888888'),
    ->
    -> (10, '0x4444444444444444444444444444444444444444');
Query OK, 15 rows affected (0.00 sec)
Records: 15  Duplicates: 0  Warnings: 0
```

```
PES2UG24CS646_PES2UG24CS916>INSERT INTO blockchain_transaction
    -> (tx_hash, amount, gas_fee, status, timestamp,
    ->  block_number, from_address, to_address)
    -> VALUES
    ->
    -> ('0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa'
    ->  5.5000000000, 0.0021000000, 'Success',
    ->  '2026-09-01 10:05:00', 100001,
    ->  '0x1111111111111111111111111111111111111111',
    ->  '0x2222222222222222222222222222222222222222'),
    ->
    -> ('0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaab'
    ->  8.2500000000, 0.0025000000, 'Success',
    ->  '2026-09-01 10:08:00', 100001,
    ->  '0x2222222222222222222222222222222222222222',
    ->  '0x3333333333333333333333333333333333333333'),
    ->
    -> ('0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb',
    ->  10.2500000000, 0.0035000000, 'Success',
    ->  '2026-09-02 11:10:00', 100002,
    ->  '0x3333333333333333333333333333333333333333',
    ->  '0x4444444444444444444444444444444444444444'),
    ->
    -> ('0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbc',
    ->  2.7500000000, 0.0015000000, 'Failed',
    ->  '2026-09-02 11:15:00', 100002,
    ->  '0x4444444444444444444444444444444444444444',
    ->  '0x5555555555555555555555555555555555555555'),
    ->
    -> ('0xccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccc',
    ->  20.0000000000, 0.0045000000, 'Success',
    ->  '2026-09-03 12:20:00', 100003,
    ->  '0x5555555555555555555555555555555555555555',
    ->  '0x6666666666666666666666666666666666666666'),
    ->
    -> ('0xccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccd',
    ->  3.7500000000, 0.0018000000, 'Pending',
    ->  '2026-09-03 12:25:00', 100003,
    ->  '0x6666666666666666666666666666666666666666',
    ->  '0x7777777777777777777777777777777777777777'),
    ->
```

```
->
-> ('0xddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddd',
->  7.2500000000, 0.0028000000, 'Pending',
->  '2026-09-04 13:30:00', 100004,
->  '0x7777777777777777777777777777777777777777',
->  '0x8888888888888888888888888888888888888888'),
->
-> ('0xddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddde',
->  15.7500000000, 0.0032000000, 'Success',
->  '2026-09-04 13:35:00', 100004,
->  '0x8888888888888888888888888888888888888888',
->  '0x9999999999999999999999999999999999999999'),
->
-> ('0xeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeee',
->  30.5000000000, 0.0050000000, 'Success',
->  '2026-09-05 14:40:00', 100005,
->  '0x9999999999999999999999999999999999999999',
->  '0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa'),
->
-> ('0xeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeef',
->  12.0000000000, 0.0025000000, 'Failed',
->  '2026-09-05 14:45:00', 100005,
->  '0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa',
->  '0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb'),
->
-> ('0xffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffff'
->  15.7500000000, 0.0032000000, 'Success',
->  '2026-09-06 15:50:00', 100006,
->  '0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb',
->  '0xccccccccccccccccccccccccccccccccccccccc'),
->
-> ('0xffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffe'
->  4.2500000000, 0.0017000000, 'Success',
->  '2026-09-06 15:55:00', 100006,
->  '0xccccccccccccccccccccccccccccccccccccccc',
->  '0x1111111111111111111111111111111111111111'),
->
```

```
-> ('0x1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef',
->  3.5000000000, 0.0012000000, 'Success',
->  '2026-09-07 16:00:00', 100007,
->  '0x1111111111111111111111111111111111111111',
->  '0x4444444444444444444444444444444444444444'),
->
-> ('0x1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdee',
->  18.0000000000, 0.0040000000, 'Success',
->  '2026-09-07 16:05:00', 100007,
->  '0x4444444444444444444444444444444444444444',
->  '0x5555555555555555555555555555555555555555'),
->
-> ('0xabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcd',
->  22.5000000000, 0.0045000000, 'Failed',
->  '2026-09-08 17:10:00', 100008,
->  '0x5555555555555555555555555555555555555555',
->  '0x6666666666666666666666666666666666666666'),
->
-> ('0xabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcdefabce',
->  6.7500000000, 0.0022000000, 'Pending',
->  '2026-09-08 17:15:00', 100008,
->  '0x6666666666666666666666666666666666666666',
->  '0x7777777777777777777777777777777777777777'),
->
-> ('0x9876543210abcdef9876543210abcdef9876543210abcdef9876543210abcdef',
->  40.0000000000, 0.0060000000, 'Success',
->  '2026-09-09 18:20:00', 100009,
->  '0x7777777777777777777777777777777777777777',
->  '0x8888888888888888888888888888888888888888'),
->
-> ('0x9876543210abcdef9876543210abcdef9876543210abcdef9876543210abcdee',
->  9.5000000000, 0.0027000000, 'Success',
->  '2026-09-09 18:25:00', 100009,
->  '0x8888888888888888888888888888888888888888',
->  '0x9999999999999999999999999999999999999999'),
->
```

```
    ->
    -> ('0xfedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbabc',
    ->  25.0000000000, 0.0048000000, 'Success',
    ->  '2026-09-10 19:30:00', 100010,
    ->  '0x9999999999999999999999999999999999999999',
    ->  '0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa'),
    ->
    -> ('0xfedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbabd',
    ->  11.2500000000, 0.0031000000, 'Failed',
    ->  '2026-09-10 19:35:00', 100010,
    ->  '0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa',
    ->  '0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb');
Query OK, 20 rows affected (0.01 sec)
Records: 20  Duplicates: 0  Warnings: 0
```

```
PES2UG24CS646_PES2UG24CS916>INSERT INTO token_transfer
    -> (token_symbol, amount, timestamp, tx_hash,
    ->  from_address, to_address)
    -> VALUES
    ->
    -> ('USDT', 1000.000000000000000000,
    ->  '2026-09-01 10:05:30',
    ->  '0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa'
    ->  '0x1111111111111111111111111111111111111111',
    ->  '0x2222222222222222222222222222222222222222'),
    ->
    -> ('USDC', 500.000000000000000000,
    ->  '2026-09-01 10:08:30',
    ->  '0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaab'
    ->  '0x2222222222222222222222222222222222222222',
    ->  '0x3333333333333333333333333333333333333333'),
    ->
    -> ('DAI', 750.000000000000000000,
    ->  '2026-09-02 11:10:30',
    ->  '0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb',
    ->  '0x3333333333333333333333333333333333333333',
    ->  '0x4444444444444444444444444444444444444444'),
    ->
    -> ('USDT', 2500.000000000000000000,
    ->  '2026-09-03 12:20:30',
    ->  '0xccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccc',
    ->  '0x5555555555555555555555555555555555555555',
    ->  '0x6666666666666666666666666666666666666666'),
    ->
    -> ('LINK', 150.000000000000000000,
    ->  '2026-09-04 13:30:30',
    ->  '0xddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddd',
    ->  '0x7777777777777777777777777777777777777777',
    ->  '0x8888888888888888888888888888888888888888'),
    ->
    -> ('ETH', 5.000000000000000000,
    ->  '2026-09-04 13:35:30',
    ->  '0xddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddde',
    ->  '0x8888888888888888888888888888888888888888',
    ->  '0x9999999999999999999999999999999999999999'),
    ->
```

```
    ->
    -> ('UNI', 300.00000000000000000,
    ->  '2026-09-05 14:40:30',
    ->  '0xeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeee',
    ->  '0x999999999999999999999999999999999999999999',
    ->  '0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa'),
    ->
    -> ('USDC', 1250.00000000000000000,
    ->  '2026-09-06 15:50:30',
    ->  '0xffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffff',
    ->  '0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb',
    ->  '0xccccccccccccccccccccccccccccccccccccccc'),
    ->
    -> ('USDT', 4000.00000000000000000,
    ->  '2026-09-07 16:00:30',
    ->  '0x1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef',
    ->  '0x11111111111111111111111111111111111111',
    ->  '0x4444444444444444444444444444444444444444'),
    ->
    -> ('DAI', 800.00000000000000000,
    ->  '2026-09-07 16:05:30',
    ->  '0x1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdee',
    ->  '0x44444444444444444444444444444444444444',
    ->  '0x5555555555555555555555555555555555555555'),
    ->
    -> ('USDT', 3500.00000000000000000,
    ->  '2026-09-09 18:20:30',
    ->  '0x9876543210abcdef9876543210abcdef9876543210abcdef9876543210abcdef',
    ->  '0x77777777777777777777777777777777777777',
    ->  '0x8888888888888888888888888888888888888888'),
    ->
    -> ('USDC', 600.00000000000000000,
    ->  '2026-09-09 18:25:30',
    ->  '0x9876543210abcdef9876543210abcdef9876543210abcdee',
    ->  '0x8888888888888888888888888888888888888888',
    ->  '0x999999999999999999999999999999999999999'),
    ->
    -> ('ETH', 8.500000000000000000,
    ->  '2026-09-10 19:30:30',
    ->  '0xfedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbabc',
    ->  '0x99999999999999999999999999999999999999',
    ->  '0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa'),
```

```
    -> ('LINK', 450.000000000000000000,
    ->  '2026-09-10 19:35:30',
    ->  '0xfedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbabd',
    ->  '0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa',
    ->  '0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb');
Query OK, 14 rows affected (0.00 sec)
Records: 14  Duplicates: 0  Warnings: 0
```

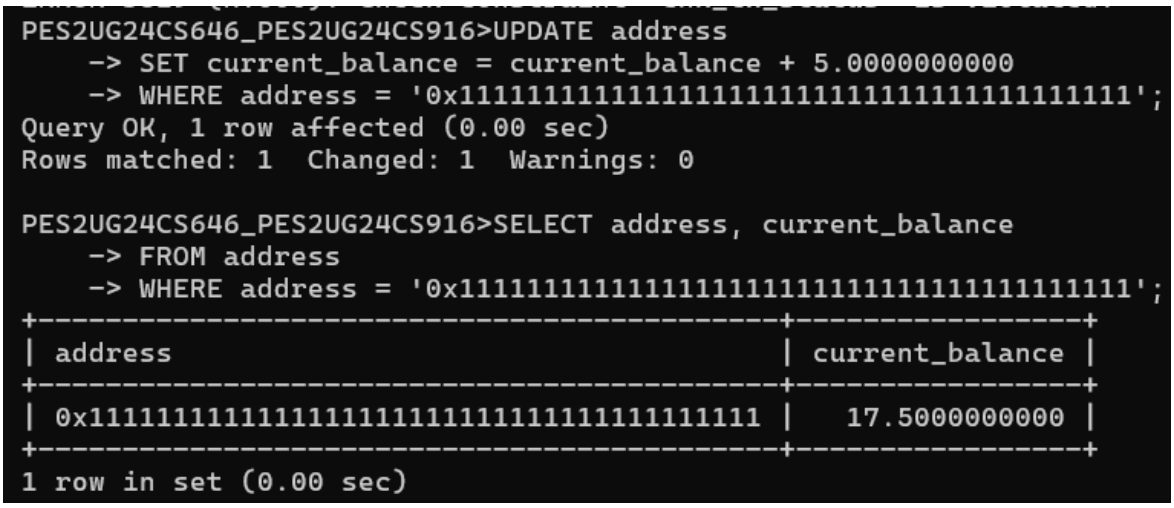
2. Constraints

```
PES2UG24CS646_PES2UG24CS916>ALTER TABLE blockchain_transaction
    -> ADD CONSTRAINT chk_tx_status
    -> CHECK (status IN ('Success', 'Failed', 'Pending'));
Query OK, 20 rows affected (0.05 sec)
Records: 20  Duplicates: 0  Warnings: 0

PES2UG24CS646_PES2UG24CS916>ALTER TABLE address
    -> ADD CONSTRAINT chk_address_type
    -> CHECK (address_type IN ('EOA', 'Contract'));
Query OK, 12 rows affected (0.03 sec)
Records: 12  Duplicates: 0  Warnings: 0
```

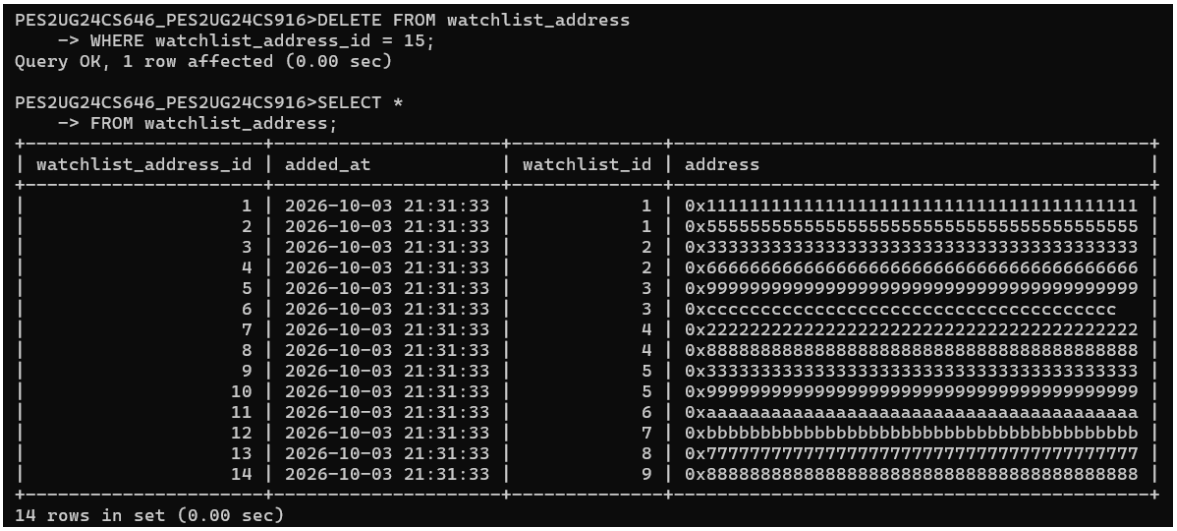
```
PES2UG24CS646_PES2UG24CS916>INSERT INTO blockchain_transaction
    -> (tx_hash, amount, gas_fee, status, timestamp,
    ->  block_number, from_address, to_address)
    -> VALUES
    -> ('0x999999999999999999999999999999999999999999999999999999999999999999',
    -> 10.0000000000,
    -> 0.0020000000,
    -> 'Invalid',
    -> '2026-10-03 20:00:00',
    -> 100001,
    -> '0x11111111111111111111111111111111111111',
    -> '0x2222222222222222222222222222222222222222');
ERROR 3819 (HY000): Check constraint 'chk_tx_status' is violated.
```

3. UPDATE and DELETE Operations



```
PES2UG24CS646_PES2UG24CS916>UPDATE address
    -> SET current_balance = current_balance + 5.0000000000
    -> WHERE address = '0x1111111111111111111111111111111111111111';
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

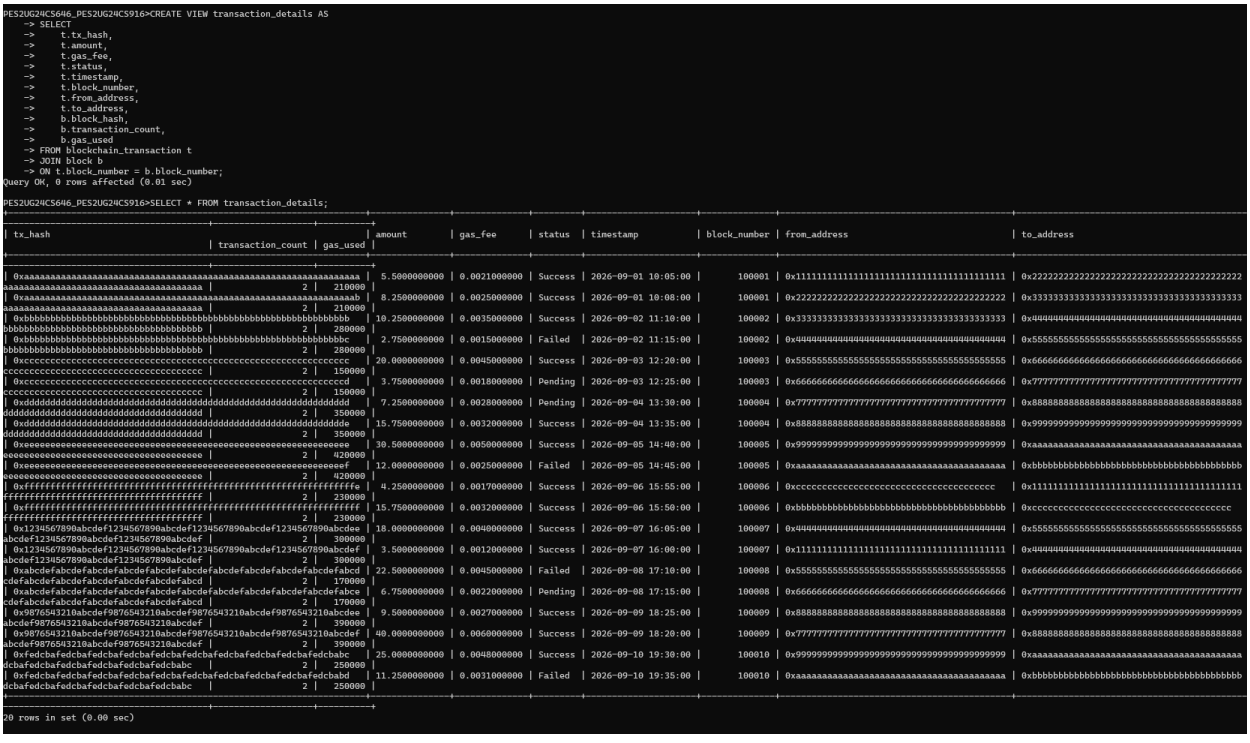
PES2UG24CS646_PES2UG24CS916>SELECT address, current_balance
    -> FROM address
    -> WHERE address = '0x1111111111111111111111111111111111111111';
+------------------------------------------+-----------------+
| address                                  | current_balance |
+------------------------------------------+-----------------+
| 0x1111111111111111111111111111111111111111 |   17.5000000000 |
+------------------------------------------+-----------------+
1 row in set (0.00 sec)
```



```
PES2UG24CS646_PES2UG24CS916>DELETE FROM watchlist_address
    -> WHERE watchlist_address_id = 15;
Query OK, 1 row affected (0.00 sec)

PES2UG24CS646_PES2UG24CS916>SELECT *
    -> FROM watchlist_address;
+----------------------+---------------------+--------------+--------------------------------------------+
| watchlist_address_id | added_at            | watchlist_id | address                                    |
+----------------------+---------------------+--------------+--------------------------------------------+
|                    1 | 2026-10-03 21:31:33 |            1 | 0x1111111111111111111111111111111111111111 |
|                    2 | 2026-10-03 21:31:33 |            1 | 0x5555555555555555555555555555555555555555 |
|                    3 | 2026-10-03 21:31:33 |            2 | 0x3333333333333333333333333333333333333333 |
|                    4 | 2026-10-03 21:31:33 |            2 | 0x6666666666666666666666666666666666666666 |
|                    5 | 2026-10-03 21:31:33 |            3 | 0x9999999999999999999999999999999999999999 |
|                    6 | 2026-10-03 21:31:33 |            3 | 0xccccccccccccccccccccccccccccccccccccccc   |
|                    7 | 2026-10-03 21:31:33 |            4 | 0x2222222222222222222222222222222222222222 |
|                    8 | 2026-10-03 21:31:33 |            4 | 0x8888888888888888888888888888888888888888 |
|                    9 | 2026-10-03 21:31:33 |            5 | 0x3333333333333333333333333333333333333333 |
|                   10 | 2026-10-03 21:31:33 |            5 | 0x9999999999999999999999999999999999999999 |
|                   11 | 2026-10-03 21:31:33 |            6 | 0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa |
|                   12 | 2026-10-03 21:31:33 |            7 | 0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb |
|                   13 | 2026-10-03 21:31:33 |            8 | 0x7777777777777777777777777777777777777777 |
|                   14 | 2026-10-03 21:31:33 |            9 | 0x8888888888888888888888888888888888888888 |
+----------------------+---------------------+--------------+--------------------------------------------+
14 rows in set (0.00 sec)
```

4. Views



```
PES2UG24CS646_PES2UG24CS916>CREATE VIEW transaction_details AS
    -> SELECT
    ->     t.tx_hash,
    ->     t.amount,
    ->     t.gas_fee,
    ->     t.status,
    ->     t.timestamp,
    ->     t.block_number,
    ->     t.from_address,
    ->     t.to_address,
    ->     b.block_hash,
    ->     b.transaction_count,
    ->     b.gas_used
    -> FROM blockchain_transaction t
    -> JOIN block b
    -> ON t.block_number = b.block_number;
Query OK, 0 rows affected (0.01 sec)

PES2UG24CS646_PES2UG24CS916>SELECT * FROM transaction_details;
+------------------------------------------------------------------+---------------+---------+--------------+---------+---------------------+--------------+------------------------------------------+------------------------------------------+-------------------+----------+
| tx_hash                                                          | amount        | gas_fee | status       | timestamp           | block_number | from_address                             | to_address                               | block_hash        | transaction_count | gas_used |
+------------------------------------------------------------------+---------------+---------+--------------+---------+---------------------+--------------+------------------------------------------+------------------------------------------+-------------------+----------+
| 0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa |  5.5000000000 | 0.0021000000 | Success | 2026-09-01 10:05:00 |       100001 | 0x1111111111111111111111111111111111111111 | 0x2222222222222222222222222222222222222222 | 0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa |                 2 |   210000 |
| 0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaab |  8.2500000000 | 0.0025000000 | Success | 2026-09-01 10:08:00 |       100001 | 0x2222222222222222222222222222222222222222 | 0x3333333333333333333333333333333333333333 | 0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa |                 2 |   210000 |
| 0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb | 10.2500000000 | 0.0035000000 | Success | 2026-09-02 11:10:00 |       100002 | 0x3333333333333333333333333333333333333333 | 0x4444444444444444444444444444444444444444 | 0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb |                 2 |   280000 |
| 0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbc |  2.7500000000 | 0.0015000000 | Failed  | 2026-09-02 11:15:00 |       100002 | 0x4444444444444444444444444444444444444444 | 0x5555555555555555555555555555555555555555 | 0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb |                 2 |   280000 |
| 0xccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccc | 20.0000000000 | 0.0045000000 | Success | 2026-09-03 12:20:00 |       100003 | 0x5555555555555555555555555555555555555555 | 0x6666666666666666666666666666666666666666 | 0xccccccccccccccccccccccccccccccccccccccccc |                 2 |   150000 |
| 0xccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccd |  3.7500000000 | 0.0018000000 | Pending | 2026-09-03 12:25:00 |       100003 | 0x6666666666666666666666666666666666666666 | 0x7777777777777777777777777777777777777777 | 0xccccccccccccccccccccccccccccccccccccccccc |                 2 |   150000 |
| 0xddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddd |  7.2500000000 | 0.0028000000 | Pending | 2026-09-04 13:30:00 |       100004 | 0x7777777777777777777777777777777777777777 | 0x8888888888888888888888888888888888888888 | 0xddddddddddddddddddddddddddddddddddddddddd |                 2 |   350000 |
| 0xddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddde | 15.7500000000 | 0.0032000000 | Success | 2026-09-04 13:35:00 |       100004 | 0x8888888888888888888888888888888888888888 | 0x9999999999999999999999999999999999999999 | 0xddddddddddddddddddddddddddddddddddddddddd |                 2 |   350000 |
| 0xeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeee | 30.5000000000 | 0.0050000000 | Success | 2026-09-05 14:40:00 |       100005 | 0x9999999999999999999999999999999999999999 | 0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa | 0xeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeee |                 2 |   420000 |
| 0xeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeef | 12.0000000000 | 0.0025000000 | Failed  | 2026-09-05 14:45:00 |       100005 | 0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa | 0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb | 0xeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeef |                 2 |   420000 |
| 0xfffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffe |  4.2500000000 | 0.0017000000 | Success | 2026-09-06 15:55:00 |       100006 | 0xccccccccccccccccccccccccccccccccccccccc   | 0x1111111111111111111111111111111111111111 | 0xfffffffffffffffffffffffffffffffffffffffe |                 2 |   230000 |
| 0xfffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffff | 17.7500000000 | 0.0032000000 | Success | 2026-09-06 15:50:00 |       100006 | 0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb | 0xccccccccccccccccccccccccccccccccccccccc   | 0xfffffffffffffffffffffffffffffffffffffffff |                 2 |   230000 |
| 0x1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdee | 18.0000000000 | 0.0040000000 | Success | 2026-09-07 16:05:00 |       100007 | 0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa | 0x5555555555555555555555555555555555555555 | 0x1234567890abcdef1234567890abcdef1234567890abcdef |                 2 |   300000 |
| 0x1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef |  3.5000000000 | 0.0012000000 | Success | 2026-09-07 16:00:00 |       100007 | 0x1111111111111111111111111111111111111111 | 0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa | 0x1234567890abcdef1234567890abcdef1234567890abcdef |                 2 |   300000 |
| 0xabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcd | 22.5000000000 | 0.0045000000 | Failed  | 2026-09-08 17:10:00 |       100008 | 0x5555555555555555555555555555555555555555 | 0x6666666666666666666666666666666666666666 | 0xabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcd |                 2 |   170000 |
| 0xabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcdefabce |  6.7500000000 | 0.0022000000 | Pending | 2026-09-08 17:15:00 |       100008 | 0x6666666666666666666666666666666666666666 | 0x7777777777777777777777777777777777777777 | 0xabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcd |                 2 |   170000 |
| 0x9876543210abcdef9876543210abcdef9876543210abcdef9876543210abcdee |  9.5000000000 | 0.0027000000 | Success | 2026-09-09 18:25:00 |       100009 | 0x8888888888888888888888888888888888888888 | 0x9999999999999999999999999999999999999999 | 0x9876543210abcdef9876543210abcdef9876543210abcdef |                 2 |   390000 |
| 0x9876543210abcdef9876543210abcdef9876543210abcdef9876543210abcdef | 40.0000000000 | 0.0060000000 | Success | 2026-09-09 18:20:00 |       100009 | 0x7777777777777777777777777777777777777777 | 0x8888888888888888888888888888888888888888 | 0x9876543210abcdef9876543210abcdef9876543210abcdef |                 2 |   390000 |
| 0xfedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbafedc | 25.0000000000 | 0.0048000000 | Success | 2026-09-10 19:30:00 |       100010 | 0x9999999999999999999999999999999999999999 | 0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa | 0xfedcbafedcbafedcbafedcbafedcbafedcbafedcbabc |                 2 |   250000 |
| 0xfedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbabd | 11.2500000000 | 0.0031000000 | Failed  | 2026-09-10 19:35:00 |       100010 | 0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa | 0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb | 0xfedcbafedcbafedcbafedcbafedcbafedcbafedcbabc |                 2 |   250000 |
+------------------------------------------------------------------+---------------+---------+--------------+---------+---------------------+--------------+------------------------------------------+------------------------------------------+-------------------+----------+
20 rows in set (0.00 sec)
```

```
PES2UG24CS646_PES2UG24CS916>CREATE VIEW wallet_activity AS
    -> SELECT
    ->     a.address,
    ->     a.address_type,
    ->     a.current_balance,
    ->     COUNT(t.tx_hash) AS transaction_count,
    ->     COALESCE(SUM(t.amount), 0) AS total_transaction_amount
    -> FROM address a
    -> LEFT JOIN blockchain_transaction t
    -> ON a.address = t.from_address
    -> GROUP BY
    ->     a.address,
    ->     a.address_type,
    ->     a.current_balance;
Query OK, 0 rows affected (0.01 sec)

PES2UG24CS646_PES2UG24CS916>SELECT * FROM wallet_activity;
+--------------------------------------------+--------------+-----------------+-------------------+--------------------------+
| address                                    | address_type | current_balance | transaction_count | total_transaction_amount |
+--------------------------------------------+--------------+-----------------+-------------------+--------------------------+
| 0x1111111111111111111111111111111111111111 | EOA          |   17.5000000000 |                 2 |             9.0000000000 |
| 0x2222222222222222222222222222222222222222 | EOA          |   25.7500000000 |                 1 |             8.2500000000 |
| 0x3333333333333333333333333333333333333333 | Contract     |  100.0000000000 |                 1 |            10.2500000000 |
| 0x4444444444444444444444444444444444444444 | EOA          |    8.2500000000 |                 2 |            20.7500000000 |
| 0x5555555555555555555555555555555555555555 | EOA          |   50.0000000000 |                 2 |            42.5000000000 |
| 0x6666666666666666666666666666666666666666 | Contract     |   75.5000000000 |                 2 |            10.5000000000 |
| 0x7777777777777777777777777777777777777777 | EOA          |   18.7500000000 |                 2 |            47.2500000000 |
| 0x8888888888888888888888888888888888888888 | EOA          |   32.0000000000 |                 2 |            25.2500000000 |
| 0x9999999999999999999999999999999999999999 | Contract     |  200.0000000000 |                 2 |            55.5000000000 |
| 0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa | EOA          |    5.5000000000 |                 2 |            23.2500000000 |
| 0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb | EOA          |   45.0000000000 |                 1 |            15.7500000000 |
| 0xcccccccccccccccccccccccccccccccccccccccc | Contract     |  150.0000000000 |                 1 |             4.2500000000 |
+--------------------------------------------+--------------+-----------------+-------------------+--------------------------+
12 rows in set (0.00 sec)
```

5. Transactions – COMMIT and ROLLBACK

```
PES2UG24CS646_PES2UG24CS916>START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

PES2UG24CS646_PES2UG24CS916>
PES2UG24CS646_PES2UG24CS916>UPDATE address
    -> SET current_balance = current_balance - 1.5000000000
    -> WHERE address =
    -> '0x1111111111111111111111111111111111111111';
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

PES2UG24CS646_PES2UG24CS916>
PES2UG24CS646_PES2UG24CS916>UPDATE address
    -> SET current_balance = current_balance + 1.5000000000
    -> WHERE address =
    -> '0x2222222222222222222222222222222222222222';
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

PES2UG24CS646_PES2UG24CS916>
PES2UG24CS646_PES2UG24CS916>COMMIT;
Query OK, 0 rows affected (0.00 sec)

PES2UG24CS646_PES2UG24CS916>SELECT address, current_balance
    -> FROM address
    -> WHERE address IN (
    -> '0x1111111111111111111111111111111111111111',
    -> '0x2222222222222222222222222222222222222222'
    -> );
+--------------------------------------------+-----------------+
| address                                    | current_balance |
+--------------------------------------------+-----------------+
| 0x1111111111111111111111111111111111111111 |   16.0000000000 |
| 0x2222222222222222222222222222222222222222 |   27.2500000000 |
+--------------------------------------------+-----------------+
2 rows in set (0.00 sec)
```

```
PES2UG24CS646_PES2UG24CS916>START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

PES2UG24CS646_PES2UG24CS916>
PES2UG24CS646_PES2UG24CS916>UPDATE address
    -> SET current_balance = current_balance - 5.0000000000
    -> WHERE address =
    -> '0x5555555555555555555555555555555555555555';
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

PES2UG24CS646_PES2UG24CS916>
PES2UG24CS646_PES2UG24CS916>UPDATE address
    -> SET current_balance = current_balance + 5.0000000000
    -> WHERE address =
    -> '0x6666666666666666666666666666666666666666';
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

PES2UG24CS646_PES2UG24CS916>
PES2UG24CS646_PES2UG24CS916>ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

PES2UG24CS646_PES2UG24CS916>SELECT address, current_balance
    -> FROM address
    -> WHERE address IN (
    -> '0x5555555555555555555555555555555555555555',
    -> '0x6666666666666666666666666666666666666666'
    -> );
+--------------------------------------------+-----------------+
| address                                    | current_balance |
+--------------------------------------------+-----------------+
| 0x5555555555555555555555555555555555555555 |   50.0000000000 |
| 0x6666666666666666666666666666666666666666 |   75.5000000000 |
+--------------------------------------------+-----------------+
2 rows in set (0.00 sec)
```

6. Stored Procedures

```
PES2UG24CS646_PES2UG24CS916>DELIMITER //
PES2UG24CS646_PES2UG24CS916>
PES2UG24CS646_PES2UG24CS916>CREATE PROCEDURE GetBlockTransactions(IN p_block_number BIGINT)
    -> BEGIN
    ->     SELECT
    ->         tx_hash,
    ->         amount,
    ->         gas_fee,
    ->         status,
    ->         timestamp,
    ->         from_address,
    ->         to_address
    ->     FROM blockchain_transaction
    ->     WHERE block_number = p_block_number;
    -> END //
Query OK, 0 rows affected (0.01 sec)

PES2UG24CS646_PES2UG24CS916>
PES2UG24CS646_PES2UG24CS916>DELIMITER ;
PES2UG24CS646_PES2UG24CS916>CALL GetBlockTransactions(100001);
+------------------------------------------------------------------+-------------+-------------+---------+---------------------+--------------------------------------------+--------------------------------------------+
| tx_hash                                                          | amount      | gas_fee     | status  | timestamp           | from_address                               | to_address                                 |
+------------------------------------------------------------------+-------------+-------------+---------+---------------------+--------------------------------------------+--------------------------------------------+
| 0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa | 5.5000000000 | 0.0021000000 | Success | 2026-09-01 10:05:00 | 0x1111111111111111111111111111111111111111 | 0x2222222222222222222222222222222222222222 |
| 0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaab | 8.2500000000 | 0.0025000000 | Success | 2026-09-01 10:08:00 | 0x2222222222222222222222222222222222222222 | 0x3333333333333333333333333333333333333333 |
+------------------------------------------------------------------+-------------+-------------+---------+---------------------+--------------------------------------------+--------------------------------------------+
2 rows in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)
```

```
PES2UG24CS646_PES2UG24CS916>DELIMITER //
PES2UG24CS646_PES2UG24CS916>
PES2UG24CS646_PES2UG24CS916>CREATE PROCEDURE GetAddressTransactions(IN p_address VARCHAR(42))
    -> BEGIN
    ->     SELECT
    ->         tx_hash,
    ->         amount,
    ->         gas_fee,
    ->         status,
    ->         timestamp,
    ->         block_number,
    ->         from_address,
    ->         to_address
    ->     FROM blockchain_transaction
    ->     WHERE from_address = p_address
    ->        OR to_address = p_address
    ->     ORDER BY timestamp DESC;
    -> END //
Query OK, 0 rows affected (0.00 sec)

PES2UG24CS646_PES2UG24CS916>
PES2UG24CS646_PES2UG24CS916>DELIMITER ;
PES2UG24CS646_PES2UG24CS916>CALL GetAddressTransactions(
    -> '0x1111111111111111111111111111111111111111'
    -> );
+------------------------------------------------------------------+-------------+-------------+---------+---------------------+--------------+--------------------------------------------+
--+
| tx_hash                                                          | amount      | gas_fee     | status  | timestamp           | block_number | from_address                               |
  |
+------------------------------------------------------------------+-------------+-------------+---------+---------------------+--------------+--------------------------------------------+
--+
| 0x1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef | 3.5000000000 | 0.0012000000 | Success | 2026-09-07 16:00:00 |       100007 | 0x1111111111111111111111111111111111111111
4 |
| 0xfffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffe | 4.2500000000 | 0.0017000000 | Success | 2026-09-06 15:55:00 |       100006 | 0xccccccccccccccccccccccccccccccccccccccc
1 |
| 0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa | 5.5000000000 | 0.0021000000 | Success | 2026-09-01 10:05:00 |       100001 | 0x1111111111111111111111111111111111111111
2 |
+------------------------------------------------------------------+-------------+-------------+---------+---------------------+--------------+--------------------------------------------+
--+
3 rows in set (0.00 sec)

Query OK, 0 rows affected (0.02 sec)
```

7. Trigger

```
PES2UG24CS646_PES2UG24CS916>DELIMITER //
PES2UG24CS646_PES2UG24CS916>
PES2UG24CS646_PES2UG24CS916>CREATE TRIGGER validate_transaction_amount
    -> BEFORE INSERT ON blockchain_transaction
    -> FOR EACH ROW
    -> BEGIN
    ->     IF NEW.amount <= 0 THEN
    ->         SIGNAL SQLSTATE '45000'
    ->         SET MESSAGE_TEXT = 'Transaction amount must be greater than zero';
    ->     END IF;
    ->
    ->     IF NEW.gas_fee < 0 THEN
    ->         SIGNAL SQLSTATE '45000'
    ->         SET MESSAGE_TEXT = 'Gas fee cannot be negative';
    ->     END IF;
    -> END //
Query OK, 0 rows affected (0.01 sec)

PES2UG24CS646_PES2UG24CS916>
PES2UG24CS646_PES2UG24CS916>DELIMITER ;
PES2UG24CS646_PES2UG24CS916>INSERT INTO blockchain_transaction
    -> (tx_hash, amount, gas_fee, status, timestamp,
    ->  block_number, from_address, to_address)
    -> VALUES
    -> (
    -> '0x8888888888888888888888888888888888888888888888888888888888888888',
    -> 0,
    -> 0.0020000000,
    -> 'Success',
    -> '2026-10-03 21:00:00',
    -> 100001,
    -> '0x1111111111111111111111111111111111111111',
    -> '0x2222222222222222222222222222222222222222'
    -> );
ERROR 1644 (45000): Transaction amount must be greater than zero
```

8. Joins

```
PES2UG24CS646_PES2UG24CS916>SELECT
    ->     t.tx_hash,
    ->     t.amount,
    ->     t.status,
    ->     t.block_number,
    ->     a1.address AS sender,
    ->     a2.address AS receiver
    -> FROM blockchain_transaction t
    -> JOIN address a1
    -> ON t.from_address = a1.address
    -> JOIN address a2
    -> ON t.to_address = a2.address;
```

tx_hash	amount	status	block_number	sender	receiver
0x1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef	3.5000000000	Success	100007	0x1111111111111111111111111111111111111111	0x4444444444444444444444444444444444444444
0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa	5.5000000000	Success	100001	0x1111111111111111111111111111111111111111	0x2222222222222222222222222222222222222222
0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaab	8.2500000000	Success	100001	0x2222222222222222222222222222222222222222	0x3333333333333333333333333333333333333333
0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb	10.2500000000	Success	100002	0x3333333333333333333333333333333333333333	0x4444444444444444444444444444444444444444
0x1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdee	18.0000000000	Success	100007	0x4444444444444444444444444444444444444444	0x5555555555555555555555555555555555555555
0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbc	2.7500000000	Failed	100002	0x4444444444444444444444444444444444444444	0x5555555555555555555555555555555555555555
0xabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcd	22.5000000000	Failed	100008	0x5555555555555555555555555555555555555555	0x6666666666666666666666666666666666666666
0xccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccc	20.0000000000	Success	100003	0x5555555555555555555555555555555555555555	0x6666666666666666666666666666666666666666
0xabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcdefabce	6.7500000000	Pending	100008	0x6666666666666666666666666666666666666666	0x7777777777777777777777777777777777777777
0xccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccd	3.7500000000	Pending	100003	0x6666666666666666666666666666666666666666	0x7777777777777777777777777777777777777777
0x9876543210abcdef9876543210abcdef9876543210abcdef9876543210abcdef	40.0000000000	Success	100009	0x7777777777777777777777777777777777777777	0x8888888888888888888888888888888888888888
0xddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddd	7.2500000000	Pending	100004	0x7777777777777777777777777777777777777777	0x8888888888888888888888888888888888888888
0x9876543210abcdef9876543210abcdef9876543210abcdef9876543210abcdee	9.5000000000	Success	100009	0x8888888888888888888888888888888888888888	0x9999999999999999999999999999999999999999
0xddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddde	15.7500000000	Success	100004	0x8888888888888888888888888888888888888888	0x9999999999999999999999999999999999999999
0xeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeee	30.5000000000	Success	100005	0x9999999999999999999999999999999999999999	0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa
0xfedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbabc	25.0000000000	Success	100010	0x9999999999999999999999999999999999999999	0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa
0xeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeef	12.0000000000	Failed	100005	0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa	0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb
0xfedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbabd	11.2500000000	Failed	100010	0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa	0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb
0xfffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffff	15.7500000000	Success	100006	0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb	0xccccccccccccccccccccccccccccccccccccccccc
0xfffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffe	4.2500000000	Success	100006	0xccccccccccccccccccccccccccccccccccccccccc	0x1111111111111111111111111111111111111111

20 rows in set (0.00 sec)

```
PES2UG24CS646_PES2UG24CS916>SELECT
    ->     w.watchlist_id,
    ->     w.name AS watchlist_name,
    ->     u.name AS user_name,
    ->     wa.address
    -> FROM watchlist w
    -> JOIN users u
    -> ON w.user_id = u.user_id
    -> JOIN watchlist_address wa
    -> ON w.watchlist_id = wa.watchlist_id;
```

watchlist_id	watchlist_name	user_name	address
1	High Value Wallets	Arjun Sharma	0x1111111111111111111111111111111111111111
1	High Value Wallets	Arjun Sharma	0x5555555555555555555555555555555555555555
2	DeFi Addresses	Rahul Kumar	0x3333333333333333333333333333333333333333
2	DeFi Addresses	Rahul Kumar	0x6666666666666666666666666666666666666666
3	Whale Tracking	Sneha Rao	0x9999999999999999999999999999999999999999
3	Whale Tracking	Sneha Rao	0xccccccccccccccccccccccccccccccccccccccc
4	Exchange Wallets	Aditya Singh	0x2222222222222222222222222222222222222222
4	Exchange Wallets	Aditya Singh	0x8888888888888888888888888888888888888888
5	Smart Contracts	Priya Nair	0x3333333333333333333333333333333333333333
5	Smart Contracts	Priya Nair	0x9999999999999999999999999999999999999999
6	Suspicious Addresses	Karan Patel	0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa
7	Token Monitoring	Ananya Das	0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb
8	Daily Transactions	Rohit Verma	0x7777777777777777777777777777777777777777
9	Large Transfers	Meera Iyer	0x8888888888888888888888888888888888888888

14 rows in set (0.00 sec)

```
PES2UG24CS646_PES2UG24CS916>SELECT
    ->     tt.transfer_id,
    ->     tt.token_symbol,
    ->     tt.amount,
    ->     tt.timestamp,
    ->     t.tx_hash,
    ->     t.status
    -> FROM token_transfer tt
    -> JOIN blockchain_transaction t
    -> ON tt.tx_hash = t.tx_hash;
```

transfer_id	token_symbol	amount	timestamp	tx_hash	status
1	USDT	1000.000000000000000000	2026-09-01 10:05:30	0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa	Success
2	USDC	500.000000000000000000	2026-09-01 10:08:30	0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaab	Success
3	DAI	750.000000000000000000	2026-09-02 11:10:30	0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb	Success
4	USDT	2500.000000000000000000	2026-09-03 12:20:30	0xccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccc	Success
5	LINK	150.000000000000000000	2026-09-04 13:30:30	0xddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddd	Pending
6	ETH	5.000000000000000000	2026-09-04 13:35:30	0xddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddde	Success
7	UNI	300.000000000000000000	2026-09-05 14:40:30	0xeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeee	Success
8	USDC	1250.000000000000000000	2026-09-06 15:50:30	0xfffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffff	Success
9	USDT	4000.000000000000000000	2026-09-07 16:00:30	0x1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef	Success
10	DAI	800.000000000000000000	2026-09-07 16:05:30	0x1234567890abcdef1234567890abcdef1234567890abcdee	Success
11	USDT	3500.000000000000000000	2026-09-09 18:20:30	0x9876543210abcdef9876543210abcdef9876543210abcdef	Success
12	USDC	600.000000000000000000	2026-09-09 18:25:30	0x9876543210abcdef9876543210abcdef9876543210abcdee	Success
13	ETH	8.500000000000000000	2026-09-10 19:30:30	0xfedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbabc	Success
14	LINK	450.000000000000000000	2026-09-10 19:35:30	0xfedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbabd	Failed

14 rows in set (0.00 sec)

9. GROUP BY and HAVING

```
PES2UG24CS646_PES2UG24CS916>SELECT
    ->     status,
    ->     COUNT(*) AS transaction_count
    -> FROM blockchain_transaction
    -> GROUP BY status;
+---------+-------------------+
| status  | transaction_count |
+---------+-------------------+
| Success |                13 |
| Failed  |                 4 |
| Pending |                 3 |
+---------+-------------------+
3 rows in set (0.00 sec)
```

```
PES2UG24CS646_PES2UG24CS916>SELECT
    ->     block_number,
    ->     COUNT(*) AS total_transactions,
    ->     SUM(amount) AS total_amount
    -> FROM blockchain_transaction
    -> GROUP BY block_number;
+--------------+--------------------+---------------+
| block_number | total_transactions | total_amount  |
+--------------+--------------------+---------------+
|       100001 |                  2 | 13.7500000000 |
|       100002 |                  2 | 13.0000000000 |
|       100003 |                  2 | 23.7500000000 |
|       100004 |                  2 | 23.0000000000 |
|       100005 |                  2 | 42.5000000000 |
|       100006 |                  2 | 20.0000000000 |
|       100007 |                  2 | 21.5000000000 |
|       100008 |                  2 | 29.2500000000 |
|       100009 |                  2 | 49.5000000000 |
|       100010 |                  2 | 36.2500000000 |
+--------------+--------------------+---------------+
10 rows in set (0.00 sec)
```

```
PES2UG24CS646_PES2UG24CS916>SELECT
    ->     token_symbol,
    ->     COUNT(*) AS transfer_count,
    ->     SUM(amount) AS total_transferred,
    ->     AVG(amount) AS average_transfer
    -> FROM token_transfer
    -> GROUP BY token_symbol;
+--------------+----------------+------------------------+----------------------------+
| token_symbol | transfer_count | total_transferred      | average_transfer           |
+--------------+----------------+------------------------+----------------------------+
| USDT         |              4 | 11000.000000000000000000 | 2750.0000000000000000000000 |
| USDC         |              3 |  2350.000000000000000000 |  783.3333333333333333333333 |
| DAI          |              2 |  1550.000000000000000000 |  775.0000000000000000000000 |
| LINK         |              2 |   600.000000000000000000 |  300.0000000000000000000000 |
| ETH          |              2 |    13.500000000000000000 |    6.7500000000000000000000 |
| UNI          |              1 |   300.000000000000000000 |  300.0000000000000000000000 |
+--------------+----------------+------------------------+----------------------------+
6 rows in set (0.00 sec)
```

```
PES2UG24CS646_PES2UG24CS916>SELECT
    ->     from_address,
    ->     COUNT(*) AS transaction_count
    -> FROM blockchain_transaction
    -> GROUP BY from_address
    -> HAVING COUNT(*) > 1;
+--------------------------------------------+-------------------+
| from_address                               | transaction_count |
+--------------------------------------------+-------------------+
| 0x1111111111111111111111111111111111111111 |                 2 |
| 0x4444444444444444444444444444444444444444 |                 2 |
| 0x5555555555555555555555555555555555555555 |                 2 |
| 0x6666666666666666666666666666666666666666 |                 2 |
| 0x7777777777777777777777777777777777777777 |                 2 |
| 0x8888888888888888888888888888888888888888 |                 2 |
| 0x9999999999999999999999999999999999999999 |                 2 |
| 0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa |                 2 |
+--------------------------------------------+-------------------+
8 rows in set (0.00 sec)
```

```
PES2UG24CS646_PES2UG24CS916>SELECT
    ->     token_symbol,
    ->     SUM(amount) AS total_amount
    -> FROM token_transfer
    -> GROUP BY token_symbol
    -> HAVING SUM(amount) > 1000;
+--------------+--------------------------+
| token_symbol | total_amount             |
+--------------+--------------------------+
| USDT         | 11000.000000000000000000 |
| USDC         |  2350.000000000000000000 |
| DAI          |  1550.000000000000000000 |
+--------------+--------------------------+
3 rows in set (0.00 sec)
```

10. ORDER BY

```
PES2UG24CS646_PES2UG24CS916>SELECT
    ->     tx_hash,
    ->     amount,
    ->     gas_fee,
    ->     status,
    ->     timestamp
    -> FROM blockchain_transaction
    -> ORDER BY amount DESC;
+------------------------------------------------------------------+---------------+--------------+---------+---------------------+
| tx_hash                                                          | amount        | gas_fee      | status  | timestamp           |
+------------------------------------------------------------------+---------------+--------------+---------+---------------------+
| 0x9876543210abcdef9876543210abcdef9876543210abcdef9876543210abcdef | 40.0000000000 | 0.0060000000 | Success | 2026-09-09 18:20:00 |
| 0xeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeee | 30.5000000000 | 0.0050000000 | Success | 2026-09-05 14:40:00 |
| 0xfedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbabc | 25.0000000000 | 0.0048000000 | Success | 2026-09-10 19:30:00 |
| 0xabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcd | 22.5000000000 | 0.0045000000 | Failed  | 2026-09-08 17:10:00 |
| 0xccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccc | 20.0000000000 | 0.0045000000 | Success | 2026-09-03 12:20:00 |
| 0x1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdee | 18.0000000000 | 0.0040000000 | Success | 2026-09-07 16:05:00 |
| 0xddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddde | 15.7500000000 | 0.0032000000 | Success | 2026-09-04 13:35:00 |
| 0xffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffff | 15.7500000000 | 0.0032000000 | Success | 2026-09-06 15:50:00 |
| 0xeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeef | 12.0000000000 | 0.0025000000 | Failed  | 2026-09-05 14:45:00 |
| 0xfedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbabd | 11.2500000000 | 0.0031000000 | Failed  | 2026-09-10 19:35:00 |
| 0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb | 10.2500000000 | 0.0035000000 | Success | 2026-09-02 11:10:00 |
| 0x9876543210abcdef9876543210abcdef9876543210abcdef9876543210abcdee |  9.5000000000 | 0.0027000000 | Success | 2026-09-09 18:25:00 |
| 0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaab |  8.2500000000 | 0.0025000000 | Success | 2026-09-01 10:08:00 |
| 0xddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddd |  7.2500000000 | 0.0028000000 | Pending | 2026-09-04 13:30:00 |
| 0xabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcdefabce |  6.7500000000 | 0.0022000000 | Pending | 2026-09-08 17:15:00 |
| 0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa |  5.5000000000 | 0.0021000000 | Success | 2026-09-01 10:05:00 |
| 0xfffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffe |  4.2500000000 | 0.0017000000 | Success | 2026-09-06 15:55:00 |
| 0xccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccd |  3.7500000000 | 0.0018000000 | Pending | 2026-09-03 12:25:00 |
| 0x1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef |  3.5000000000 | 0.0012000000 | Success | 2026-09-07 16:00:00 |
| 0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbc |  2.7500000000 | 0.0015000000 | Failed  | 2026-09-02 11:15:00 |
+------------------------------------------------------------------+---------------+--------------+---------+---------------------+
20 rows in set (0.00 sec)
```

```
PES2UG24CS646_PES2UG24CS916>SELECT
    ->     tx_hash,
    ->     amount,
    ->     status,
    ->     timestamp
    -> FROM blockchain_transaction
    -> ORDER BY timestamp DESC;
+------------------------------------------------------------------+---------------+---------+---------------------+
| tx_hash                                                          | amount        | status  | timestamp           |
+------------------------------------------------------------------+---------------+---------+---------------------+
| 0xfedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbabd |  11.2500000000 | Failed  | 2026-09-10 19:35:00 |
| 0xfedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbabc |  25.0000000000 | Success | 2026-09-10 19:30:00 |
| 0x9876543210abcdef9876543210abcdef9876543210abcdef9876543210abcdee |   9.5000000000 | Success | 2026-09-09 18:25:00 |
| 0x9876543210abcdef9876543210abcdef9876543210abcdef9876543210abcdef |  40.0000000000 | Success | 2026-09-09 18:20:00 |
| 0xabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcdefabce |   6.7500000000 | Pending | 2026-09-08 17:15:00 |
| 0xabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcd |  22.5000000000 | Failed  | 2026-09-08 17:10:00 |
| 0x1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdee |  18.0000000000 | Success | 2026-09-07 16:05:00 |
| 0x1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef |   3.5000000000 | Success | 2026-09-07 16:00:00 |
| 0xfffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffe |   4.2500000000 | Success | 2026-09-06 15:55:00 |
| 0xffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffff |  15.7500000000 | Success | 2026-09-06 15:50:00 |
| 0xeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeef |  12.0000000000 | Failed  | 2026-09-05 14:45:00 |
| 0xeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeee |  30.5000000000 | Success | 2026-09-05 14:40:00 |
| 0xddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddde |  15.7500000000 | Success | 2026-09-04 13:35:00 |
| 0xddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddd |   7.2500000000 | Pending | 2026-09-04 13:30:00 |
| 0xccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccd |   3.7500000000 | Pending | 2026-09-03 12:25:00 |
| 0xccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccc |  20.0000000000 | Success | 2026-09-03 12:20:00 |
| 0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbc |   2.7500000000 | Failed  | 2026-09-02 11:15:00 |
| 0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb |  10.2500000000 | Success | 2026-09-02 11:10:00 |
| 0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaab |   8.2500000000 | Success | 2026-09-01 10:08:00 |
| 0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa |   5.5000000000 | Success | 2026-09-01 10:05:00 |
+------------------------------------------------------------------+---------------+---------+---------------------+
20 rows in set (0.00 sec)
```

11. Aggregate Functions

```
PES2UG24CS646_PES2UG24CS916>SELECT
    ->     COUNT(*) AS total_transactions,
    ->     SUM(amount) AS total_amount,
    ->     AVG(amount) AS average_amount,
    ->     MIN(amount) AS minimum_amount,
    ->     MAX(amount) AS maximum_amount
    -> FROM blockchain_transaction;
+--------------------+----------------+--------------------+----------------+----------------+
| total_transactions | total_amount   | average_amount     | minimum_amount | maximum_amount |
+--------------------+----------------+--------------------+----------------+----------------+
|                 20 | 272.5000000000 | 13.62500000000000 |   2.7500000000 |  40.0000000000 |
+--------------------+----------------+--------------------+----------------+----------------+
1 row in set (0.00 sec)
```

```
PES2UG24CS646_PES2UG24CS916>SELECT
    ->     COUNT(*) AS total_transactions,
    ->     SUM(gas_fee) AS total_gas_fee,
    ->     AVG(gas_fee) AS average_gas_fee,
    ->     MIN(gas_fee) AS minimum_gas_fee,
    ->     MAX(gas_fee) AS maximum_gas_fee
    -> FROM blockchain_transaction;
+--------------------+---------------+-------------------+-----------------+-----------------+
| total_transactions | total_gas_fee | average_gas_fee   | minimum_gas_fee | maximum_gas_fee |
+--------------------+---------------+-------------------+-----------------+-----------------+
|                 20 |  0.0628000000 | 0.00314000000000 |    0.0012000000 |    0.0060000000 |
+--------------------+---------------+-------------------+-----------------+-----------------+
1 row in set (0.00 sec)
```

12. Subqueries

```
PES2UG24CS646_PES2UG24CS916>SELECT DISTINCT address
    -> FROM address
    -> WHERE address IN
    -> (
    ->     SELECT from_address
    ->     FROM blockchain_transaction
    ->     WHERE amount >
    ->     (
    ->         SELECT AVG(amount)
    ->         FROM blockchain_transaction
    ->     )
    -> );
+--------------------------------------------+
| address                                    |
+--------------------------------------------+
| 0x4444444444444444444444444444444444444444 |
| 0x5555555555555555555555555555555555555555 |
| 0x7777777777777777777777777777777777777777 |
| 0x8888888888888888888888888888888888888888 |
| 0x9999999999999999999999999999999999999999 |
| 0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb |
+--------------------------------------------+
6 rows in set (0.00 sec)
```

```
PES2UG24CS646_PES2UG24CS916>SELECT
    ->     address,
    ->     current_balance
    -> FROM address
    -> WHERE current_balance >
    -> (
    ->     SELECT AVG(current_balance)
    ->     FROM address
    -> );
+--------------------------------------------+-----------------+
| address                                    | current_balance |
+--------------------------------------------+-----------------+
| 0x3333333333333333333333333333333333333333 |  100.0000000000 |
| 0x6666666666666666666666666666666666666666 |   75.5000000000 |
| 0x9999999999999999999999999999999999999999 |  200.0000000000 |
| 0xcccccccccccccccccccccccccccccccccccccccc  |  150.0000000000 |
+--------------------------------------------+-----------------+
4 rows in set (0.00 sec)
```

```
PES2UG24CS646_PES2UG24CS916>SELECT
    ->     block_number,
    ->     transaction_count
    -> FROM block
    -> WHERE transaction_count >
    -> (
    ->     SELECT AVG(transaction_count)
    ->     FROM block
    -> );
Empty set (0.00 sec)
```

```
PES2UG24CS646_PES2UG24CS916>SELECT
    ->     tx_hash,
    ->     amount,
    ->     from_address
    -> FROM blockchain_transaction
    -> WHERE from_address IN
    -> (
    ->     SELECT address
    ->     FROM address
    ->     WHERE current_balance >
    ->     (
    ->         SELECT AVG(current_balance)
    ->         FROM address
    ->     )
    -> );
+------------------------------------------------------------------+---------------+--------------------------------------------+
| tx_hash                                                          | amount        | from_address                               |
+------------------------------------------------------------------+---------------+--------------------------------------------+
| 0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb | 10.2500000000 | 0x3333333333333333333333333333333333333333 |
| 0xabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcdefabce |  6.7500000000 | 0x6666666666666666666666666666666666666666 |
| 0xccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccd |  3.7500000000 | 0x6666666666666666666666666666666666666666 |
| 0xeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeee | 30.5000000000 | 0x9999999999999999999999999999999999999999 |
| 0xfedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbabc | 25.0000000000 | 0x9999999999999999999999999999999999999999 |
| 0xfffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffe |  4.2500000000 | 0xccccccccccccccccccccccccccccccccccccccc   |
+------------------------------------------------------------------+---------------+--------------------------------------------+
6 rows in set (0.00 sec)
```

13. EXISTS and NOT EXISTS

```
PES2UG24CS646_PES2UG24CS916>SELECT
    ->     a.address,
    ->     a.current_balance
    -> FROM address a
    -> WHERE EXISTS
    -> (
    ->     SELECT 1
    ->     FROM blockchain_transaction t
    ->     WHERE t.from_address = a.address
    -> );
+--------------------------------------------+-----------------+
| address                                    | current_balance |
+--------------------------------------------+-----------------+
| 0x1111111111111111111111111111111111111111 |   16.0000000000 |
| 0x2222222222222222222222222222222222222222 |   27.2500000000 |
| 0x3333333333333333333333333333333333333333 |  100.0000000000 |
| 0x4444444444444444444444444444444444444444 |    8.2500000000 |
| 0x5555555555555555555555555555555555555555 |   50.0000000000 |
| 0x6666666666666666666666666666666666666666 |   75.5000000000 |
| 0x7777777777777777777777777777777777777777 |   18.7500000000 |
| 0x8888888888888888888888888888888888888888 |   32.0000000000 |
| 0x9999999999999999999999999999999999999999 |  200.0000000000 |
| 0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa |    5.5000000000 |
| 0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb |   45.0000000000 |
| 0xccccccccccccccccccccccccccccccccccccccc   |  150.0000000000 |
+--------------------------------------------+-----------------+
12 rows in set (0.00 sec)
```

```
PES2UG24CS646_PES2UG24CS916>SELECT
    ->     a.address,
    ->     a.current_balance
    -> FROM address a
    -> WHERE NOT EXISTS
    -> (
    ->     SELECT 1
    ->     FROM blockchain_transaction t
    ->     WHERE t.from_address = a.address
    ->        OR t.to_address = a.address
    -> );
Empty set (0.00 sec)
```

14. ANALYTICS

```
PES2UG24CS646_PES2UG24CS916>SELECT
    ->     a.address,
    ->     COUNT(t.tx_hash) AS transaction_count
    -> FROM address a
    -> JOIN blockchain_transaction t
    -> ON a.address = t.from_address
    -> GROUP BY a.address
    -> ORDER BY transaction_count DESC;
+--------------------------------------------+-------------------+
| address                                    | transaction_count |
+--------------------------------------------+-------------------+
| 0x1111111111111111111111111111111111111111 |                 2 |
| 0x4444444444444444444444444444444444444444 |                 2 |
| 0x5555555555555555555555555555555555555555 |                 2 |
| 0x6666666666666666666666666666666666666666 |                 2 |
| 0x7777777777777777777777777777777777777777 |                 2 |
| 0x8888888888888888888888888888888888888888 |                 2 |
| 0x9999999999999999999999999999999999999999 |                 2 |
| 0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa |                 2 |
| 0x2222222222222222222222222222222222222222 |                 1 |
| 0x3333333333333333333333333333333333333333 |                 1 |
| 0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb |                 1 |
| 0xccccccccccccccccccccccccccccccccccccccc   |                 1 |
+--------------------------------------------+-------------------+
12 rows in set (0.00 sec)
```

```
PES2UG24CS646_PES2UG24CS916>SELECT
    ->     token_symbol,
    ->     SUM(amount) AS total_volume,
    ->     AVG(amount) AS average_transfer,
    ->     COUNT(*) AS number_of_transfers
    -> FROM token_transfer
    -> GROUP BY token_symbol
    -> ORDER BY total_volume DESC;
+--------------+----------------------+----------------------------+---------------------
| token_symbol | total_volume         | average_transfer           | number_of_transfers
+--------------+----------------------+----------------------------+---------------------
| USDT         | 11000.000000000000000000 | 2750.000000000000000000000 |                   4
| USDC         |  2350.000000000000000000 |  783.333333333333333333333 |                   3
| DAI          |  1550.000000000000000000 |  775.000000000000000000000 |                   2
| LINK         |   600.000000000000000000 |  300.000000000000000000000 |                   2
| UNI          |   300.000000000000000000 |  300.000000000000000000000 |                   1
| ETH          |    13.500000000000000000 |    6.750000000000000000000 |                   2
+--------------+----------------------+----------------------------+---------------------
6 rows in set (0.00 sec)
```

```
PES2UG24CS646_PES2UG24CS916>SELECT
    ->     block_number,
    ->     transaction_count,
    ->     gas_used,
    ->     ROUND(
    ->         gas_used / NULLIF(transaction_count, 0),
    ->         2
    ->     ) AS avg_gas_per_transaction
    -> FROM block
    -> ORDER BY gas_used DESC;
+--------------+-------------------+----------+-------------------------+
| block_number | transaction_count | gas_used | avg_gas_per_transaction |
+--------------+-------------------+----------+-------------------------+
|       100005 |                 2 |   420000 |               210000.00 |
|       100009 |                 2 |   390000 |               195000.00 |
|       100004 |                 2 |   350000 |               175000.00 |
|       100007 |                 2 |   300000 |               150000.00 |
|       100002 |                 2 |   280000 |               140000.00 |
|       100010 |                 2 |   250000 |               125000.00 |
|       100006 |                 2 |   230000 |               115000.00 |
|       100001 |                 2 |   210000 |               105000.00 |
|       100008 |                 2 |   170000 |                85000.00 |
|       100003 |                 2 |   150000 |                75000.00 |
+--------------+-------------------+----------+-------------------------+
10 rows in set (0.00 sec)
```